

Matthew Pham

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Technical Skills

Programming Languages: Bash, C, HTML/CSS, Java, JavaScript, Python

Libraries/Framework: Matplotlib, Numpy, Pandas, React, Scikit-learn, Scrapy

Operating Systems: Microsoft Windows 10/11, MacOS, Linux

Transferable Skills

- Strong ability to analyze problems, debug code, and find efficient solutions achieved from coursework and self-learning projects.
- Effective collaboration skills gained from team-based assignments and group coding projects.
- Good time management skills developed from balancing multiple school assignments, personal projects, and extracurricular activities.

Projects

National Mens Team Predictor

July 2026

- Built an international football match predictor from scratch in Python, processing 49,000+ historical matches (1872–2026) into a validated win/draw/loss system.
- Implemented an Elo rating engine from first principles, including a home-advantage adjustment, to rank 300+ national teams by relative strength using only match results.
- Trained a Poisson regression model to predict expected goals from Elo rating differentials.
- Built a Monte Carlo simulation to convert expected-goals predictions into win/draw/loss probabilities.
- Tools Used: Python, Pandas, Numpy, Scikit-learn

Biomedical Computing Project

Sept. 2025 - Dec. 2025

- Developed an unsupervised learning pipeline to cluster cell types from 3D DNA-PAINT protein localization data using MATLAB, t-SNE, and k-means.
- Implemented protein surface-distance and colocalization analysis functions to determine protein localization patterns.
- Communicated technical results and design decisions through weekly meetings, written reports, and figures.
- Tools Used: MATLAB, Git, LaTeX

SFU Parking Mayhem

Jan. 2025 - Apr. 2025

- Programmed the movement of the player's character, board boundary, and collision between objects on the board.
- Refactored methods to make code testable and increase readability of code.
- Created test cases for character movement, collision between entities, functionality of the player's score.
- Tested and optimized running time using JaCoCo to increase efficiency of the game.
- Tools Used: Java, JavaDoc, JaCoCo, Git

Interests

EDM/Music Festivals, Hikes, Soccer, Running, Learn new tools and technologies.

Education

Simon Fraser University, B.Sc in Computing Science

Sept. 2022 - Present

- **Relevant Coursework:** Computer Architecture, Data Structures, Algorithms, Computational Theory.